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1 Introduction

The report presents the key findings and analysis from the notebook work following the course COSC2753 - Machine Learning Assignment 1 requirements. The principal objective is to explore different machine learning techniques, particularly regression, to address a practical issue. The dataset used for this assignment contains the target variable - life expectancy in years across various countries - along with other features representing a variety of life domains including health, disease immunization, economy, and demography.

2 Objective

The ultimate goal is to identify the most effective regression model in predicting life expectancy using the provided dataset. To achieve this, the process involves: - Conducting exploratory data analysis (EDA) to obtain deeper insights into feature distributions, and interdependencies, which contributes to selecting appropriate feature engineering, scaling methods, model selection, and evaluation metrics. - Preprocessing, including selection, transformation, and possible creation of new features based on key insights from EDA. - Building and comparing multiple regression models to assess their performance, using appropriate evaluation metrics.

3 Preprocessing and EDA

3.1 Missing value handling

The original dataset contains a small proportion of missing values, in which none of them is contextual data, like "Year". Therefore, instead of removing these entries - which could result in the loss of valuable information - a more appropriate approach is to apply imputation. The chosen method is K-Nearest Neighbors (KNN) imputer, which estimates missing values based on neighboring data points. Given the low percentage of missing data, this method is expected to have minimal impact on overall model performance. Furthermore, due to the presence of temporal variable, KNN algorithm will help preserve the chronological nature of data.

3.2 Observations of univariate analysis:

- Imbalance in categorical variable: There is a major imbalance in the feature "Status", indicating "developed" has substantially fewer samples than "developing". This imbalance may lead to biased model performance, potentially resulting in lower prediction accuracy for developed countries.

- Skewed distribution: The majority of features are highly skewed, signifying a violation of the normality assumption of linear regression, which requires transforming steps to symmetrize the distribution.
- Outlier presence: Outliers are prevalent across all features, suggesting the need to apply robust scaling or outlier handling techniques to minimize the impact of extreme outliers, and the risk of overfitting.

3.3 Observations of multivariate analysis:

- Multicollinearity in Mortality Features: Box plots of “AdultMortality”, “AdultMortality-Male”, and “AdultMortality-Female” indicate a high degree of correlation among these features, suggesting the presence of multicollinearity condition. This observation is further supported by the correlation matrix heatmap. To reduce redundancy, “AdultMortality-Male” and “AdultMortality-Female” will be removed, keeping only “AdultMortality”, which represents overall mortality across genders.
- The Histogram of Polio and Diphtheria immunization coverage among 1-year-olds also implies a notable correlation. The correlation coefficient between them is 0.69, which, while moderately high, remains below the common threshold of 0.8 [1] for multicollinearity concerns. Therefore, both features will be retained without combining or removal.
- Life Expectancy Trends by Country: The line plot titled “Life Expectancy Trends of Selected Countries” illustrates the temporal life expectancy patterns of six countries. It can be inferred that each line fluctuates with minimal differences between two consecutive years, with countries naturally grouped into two clusters - those with life expectancy above 70 years and those below.
- Overall Trend in Life Expectancy: The plot “Mean of Life Expectancy Throughout the Years” demonstrates an apparent upward trend, indicating general improvements in global life expectancy over time.
- Life Expectancy in different development status: The “Boxplot of Life Expectancy grouped by status” indicates that developed countries have significantly higher life expectancy compared to developing countries, emphasizing how socioeconomic development contributes to public health.
- Relationship between features and target variable: Insights from the pair plot indicate that most features exhibit weak linear relationships with the target variable. Therefore, non-linear models, such as tree-based algorithms or neighbor-based methods, may perform better than linear models in capturing the underlying patterns in the data [2].
- High correlations between two features: The pair plot of socioeconomic indicators reveals some notable linear relationships. In particular, “GDP” and “PercentageExpenditure”, as well as “IncomeCompositionOfResources” and “Schooling”, demonstrate strong positive correlations, with Pearson coefficients of 0.92 and 0.75, respectively.
- The moderate correlation between Income Composition and Schooling aligns with expectations, as both are components of the Human Development Index (HDI) [3]. This collinearity reflects the intuitive relationship that higher education levels are generally associated with higher income. However, the coefficient remains within acceptable bounds and does not necessitate transformation or removal of either feature.
- Conversely, the extremely high correlation between “GDP” and “PercentageExpenditure” suggests a strong linear dependency, indicating that countries with larger economies tend to allocate a larger proportion of resources to healthcare. Due to the strength of this relationship, additional steps may be needed to address multicollinearity.

- Observation of the pair plot of health indicators shows that "Thinness1-19years" and "Thinness5-9years" have an extremely strong linear relationship, which can be deducted from the fact that they represent the same health metric of different age groups. Given the high degree of similarity, retaining both variables may introduce redundancy and multicollinearity into the model. To preserve their informational value while addressing their linear dependency, merging them into a single feature is the ideal approach.
- The correlation between "Under5LS" and "SLS" is also exceptionally high, with a coefficient of 0.99, indicating that they capture nearly identical information. This strong linear relationship is expected, as both features represent the proportion of short life spans within a population. However, "SLS" is presumed to include a broader age range beyond children under five, potentially offering a more comprehensive view of early mortality. Therefore, retaining "SLS" and removing "Under5LS" is an appropriate choice to reduce redundancy while preserving the information rather than combining them.
- Evaluation metrics selection: The high presence of outliers suggests careful handling to prevent potential overfitting during model training. Mean Absolute Error (MAE) is selected as the main evaluation metric, as it treats all errors equally and is less sensitive to large deviations compared to Mean Squared Error (MSE). This makes MAE more robust in the presence of outliers. Additionally, the R^2 score will be used as a secondary metric together with MSE, with attention to unusually high R^2 values, which may be a signal of overfitting rather than true generalization.

3.4 Scaling and transformation

- To address the issues posed by outliers and skewed distribution, Robust Scaler and Quantile Transformer will be applied and incorporated into the pipeline to ensure that the model training process remains resilient to extreme values and achieves a more uniform data distribution.

3.5 Feature Engineering:

- Create feature "Thinness5-19years": The new feature is created by taking the average of both thinness indicators.
- Drop less informative feature: "PercentageExpenditure" will be excluded from the features set while retaining "GDP" as "GDP" is a broader representation of different life domains, apart from healthcare expenditure. Although healthcare is a contributing factor to life expectancy, it is not the sole determinant [4].
- Exclude features: All redundant features will be dropped including ID, adult mortality of genders, original thinness features, and mortality under the age of 5.

4 Performance analysis of models: (see appendix for evaluation metrics summary)

- Model selection: A diverse set of models was explored, including both linear regression models (Linear Regression, Lasso, Ridge, Elastic Net) and non-linear models (Random Forest, Gradient Boosting, HistGradientBoosting, Extra Trees, Decision Tree, AdaBoost with Decision Tree estimator, KNN, XGBoost, and Support Vector Regression (SVR)). Each model underwent hyperparameter tuning to optimize performance. The top-performing models were further selected for ensembling. Although EDA indicates that non-linear models have greater potential in capturing complex relationships within the data, both linear and non-linear models will be thoroughly assessed. This comprehensive approach ensures a more generalizable and balanced conclusion regarding model characteristic, hence, provide valuable insight for modeling strategies for similar tasks in the future.

- Hyperparameter selection: The parameter grids will be manually changed across different attempts to achieve the optimal outcome with concise parameter ranges to save the resources. Numerical parameter will be initially large, then progressively narrowing down based on the output of the previous attempt.
- Base models comparison: Initial comparisons reveal that non-linear models consistently outperform linear models including manually tuned regularization models, confirming the non-linear nature of the dataset as observed in EDA. Among them, tree-based methods generally show strong performance, while the KNN model underperforms, likely due to the high-dimensional data with noisy outliers [5]. Remarkably, Decision Tree demonstrates weaker performance when evaluated against more advanced ensemble-based tree models, indicating that model complexity and ensembling play crucial roles in enhancing predictive capabilities. This suggests that further ensemble models combining multiple tree-based approaches may yield even better prediction than any individual model.
- Importance of Cross-Validation: The outputs of tree-based models are not only distinct from each other but also show noticeable variation across multiple training attempts. This fluctuation is likely due to differences in the training and validation splits used during each run. Such inconsistency suggests potential vulnerability when these models are applied to unseen data. Therefore, cross-validation becomes essential—even for baseline models—to ensure stability and reliability.
- Manually tuning regularization parameter: By plotting the trend of MAE score across a predefined range of parameter values, the optimal parameter and potential range for further tuning can be identified. At the same time, to ensure both efficiency and effectiveness, the parameter grid was kept concise by eliminating redundant values. This strategy reduces computational overhead while still capturing the regularization strength's impact on the model.
- The introduction of polynomial features has considerably improved linear regression models by capturing underlying linear patterns. The degree of polynomial features was limited due to high computational cost when implemented in a dataset with numerous columns, which may have constrained the full potential of this technique. Additionally, applying polynomial feature expansion to tree-based models is not ideal, as this type of model inherently captures non-linear interactions, resulting in unnecessary computational complexity without benefitting the models.
- Hyperparameter tuning using Randomized Search with scoring and cross-validation has generally led to notable improvements in the predictive accuracy of non-linear models. Analysis based on MAE scores indicates that most non-linear algorithms benefited from this process, showcasing higher evaluation score. However, Random Forest and Extra Trees did not gain benefit from the tuning, suggesting that their default configurations may already be close to optimal or that the parameter grids were not impactful. It is also important to note that inconsistent outputs were observed across different tuning attempts. The phenomenon could be resulted from the use of random integers in parameter grids, which may lead to different parameter combinations being evaluated each time. Using the same approach with manual tuning for regularization model, numerical parameter range can be limited to closer to the obtained optimal value.
- Room for improvement in real-world application of hyperparameter tuning: Due to resource constraint, optimal hyperparameter estimation technique can not be implemented in this

assignment, resulting in not being able to identify the best configuration of models. Grid Search can be employed instead of Randomized Search to exhaustively explore a wider range of parameter combinations. In addition, defining a more targeted and refined parameter grid can help ensure more consistent and reliable outcomes, reducing variability and enhancing the reproducibility of model performance across runs.

- Regularized models' poor performance on unseen data: Manual K-Fold cross-validation was implemented on regularized linear models, with and without polynomial features, to try all possible configuration on different subsets of training data. The techniques provides a more general and robust assessment of the models, where their performance on unseen data can be observed. The implementation resulted in noticeably higher average error metrics in comparison to training without cross-validation. This outcome suggests a considerable variation in performance across folds, which reflects a lack of consistency in the models' predictive capabilities. Consequently, this implies that the regularized models may not generalize well to unseen data, indicating potential overfitting from the training data.
- Resources and approach identification: The computational costs associated with hyperparameter tuning for both linear and non-linear models were found to be relatively similar. However, the substantial difference in their predictive performance highlights the importance of conducting a thorough EDA. A well-executed EDA can provide valuable insights into the underlying patterns and relationships within the dataset, which can guide the selection of the most suitable modeling approach. This step is essential to ensure optimal model performance while maintaining computational efficiency.
- Voting Regressor does not outperform the optimal standalone model: Five best tree-based models are utilized as estimators for Voting Regressor. Despite the outputed MAE being comparative, it did not surpass the base Extra Tree model. This result is expected as the Voting Regressor averages all estimators, which can sometimes dilute the strengths of strong models when ensembled with weaker ones.
- Stacking Regressor achieved the highest evaluation metrics among all implemented models: Using the same best five models, this ensemble technique was able to output such performance by collectively stacking all estimators, instead of taking the average, which maximized all the model's strength. However, as mentioned above, the model is built based on inconsistent tree-based model, resulting in the variablity of performance across training attempts and possible different estimators' parameters each time. Despite the advantage, it will remained chosen as the optimal model as the problem can be solved by allocating more resource and stacking generalization can guarantee better performance than standalone models.

5 Conclusion

- When model accuracy is the highest priority, ensemble models are highly recommended, as they consistently deliver superior evaluation metrics. In such cases, exhaustive hyperparameter tuning methods yields more consistent results.
- However, if computational cost is a limiting factor, a thorough EDA can provide guidance in narrowing down the model selection to only those with the most potential. In addition to strategic parameter choice, it is still possible to achieve relatively high-performing models.
- The selection of evaluation metrics and loss functions should be informed by insights gathered during EDA, ensuring alignment with the underlying data characteristics and project goals.
- Incorporating domain knowledge is highly beneficial for identifying and eliminating redundant features, which not only simplifies the model but can also improve its performance.

6 References

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7 Appendix

Model	MAE	MSE	R^2
LinearRegression	3.9666	25.0812	0.6898
RandomForest	2.0673	7.2535	0.9103
GradientBoosting	2.2764	8.1519	0.8992
HistGradientBoosting	2.1428	7.4188	0.9082
DecisionTree	3.0764	15.8567	0.8039
ExtraTrees	1.9475	6.3520	0.9214
KNN	2.7231	13.0332	0.8388
AdaBoost	2.6867	11.0013	0.8639
XGBoost	2.3206	8.6660	0.8928
Lasso	3.9663	25.0714	0.6899
Ridge	3.9665	25.0788	0.6898
ElasticNet	3.9640	24.8050	0.6932
SVR	2.3827	10.3652	0.8718
Random Forest (Tuned)	2.0854	7.4333	0.9081
GradientBoosting (Tuned)	2.0675	6.9930	0.9135
HistGradientBoosting (Tuned)	2.1316	7.6208	0.9057
DecisionTree (Tuned)	2.5967	11.4246	0.8587
ExtraTrees (Tuned)	2.2258	8.1968	0.8986
KNN (Tuned)	2.2666	8.6791	0.8927

Model	MAE	MSE	R^2
AdaBoost (Tuned)	2.7380	11.5040	0.8577
XGBoost (Tuned)	2.0835	7.1219	0.9119
Polynomial Lasso	2.6134	12.0818	0.8506
Polynomial Ridge	2.6596	12.4163	0.8464
Polynomial ElasticNet	2.6095	12.2266	0.8488
Polynomial SVR	2.4668	10.9872	0.8641
Lasso with CV	4.0095	27.0354	0.7069
Ridge with CV	4.0108	27.0649	0.7066
ElasticNet with CV	4.0095	27.0354	0.7069
SVR with CV	2.7255	13.2409	0.8561
Polynomial Lasso with CV	2.8840	14.8436	0.8390
Polynomial Ridge with CV	2.9281	15.5234	0.8311
Polynomial ElasticNet with CV	2.8796	14.8548	0.8388
Polynomial SVR with CV	2.8597	14.5219	0.8423
Voting Regressor	1.9932	6.5255	0.9193
Stacking Regressor	1.9393	6.1052	0.9245